Course Name: Fundamentals of Programming & Data Science	**Course Code:** CMPE-112L
Assignment Type: Complex Engineering Problem	**Dated: 09-May-2023**
Semester: 2nd	**Session: 2022**
Lab/Project/Assignment #: 1	**CLOs to be covered: CLO 2**
Lab Title: Encryption and Decryption Tool	**Teacher Name:** Afeef Obaid

Complex Engineering Problem (CEP):

No.	Attribute	Details	PLOs Covered
WP1	Depth of Knowledge required	This problem requires deep knowledge of dictionaries, random functions and string manipulation in python.	**1**
WP2	Range of conflicting requirements	This problem involves conflicting requirements, as we can make a very complex encryption and decryption algorithm but its processing time, efficiency and code complexity will increase on the other hand.	**3**
WP7	Interdependence	This problem has interdependence as algorithm for decryption depends upon the algorithm which we apply for encryption.	**4**

CEP Description:

Write the python program that takes a sentence from the user and do its Encryption or decryption using dictionary

Rules for encryption:

1. Reverse the word

2. Replace each letter of the word with some random letter

3. Add three random alphabets before and after each word

Rules for Decryption:

Decryption can be done by reversing the encryption steps.